

Dually Flat Affine Geometry of the Probability Simplex Bases on Kaniadakis' Logarithm for Applications to Compositional Data Statistics

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Abstract: We use Kaniadakis' logarithm to mimic the standard construction of the affine statistical manifold. The parallel transport duality is conserved with an adaptation of the set-up consisting in the introduction of the escort probability functions and a new bilinear form on fibres. The use of exponential charts requires computations to be implemented.

Keywords: Information geometry; Compositional data; Statistical bundle; Kaniadakis entropy.

1. Introduction

1.1. Kaniadakis' logarithm

We focus on a special case of Kaniadakis' logarithm [1,2]. In the suggestive formalism introduced by Naudts [3],

$$\log_K(x) = \frac{1}{2} \left(x - \frac{1}{x} \right) = \int_1^x \frac{du}{A(u)}, \quad \text{with} \quad A(u) = \frac{2u^2}{1+u^2}, \quad (1)$$

$$\exp_K(y) = \log_K^{-1} y = y + \sqrt{1+y^2} = \exp \int_0^y \frac{dv}{\sqrt{1+v^2}}. \quad (2)$$

See ??.

The graph of $y = \log_K(x)$ is a semi-algebraic function,

$$x^2 - 2xy - 1 = 0, \quad x > 0.$$

The equation reduces any polynomial in y and $x = \exp_K(y)$, for example,

$$\begin{aligned} \exp_K(y)^2 &= 2y \exp_K(y) + 1, \\ \exp_K(y)^3 &= \exp_K(y)(2y \exp_K(y) + 1) = \dots, \end{aligned}$$

and so on. This is an algebraic feature, cf [4].

We list below the main known properties of the K-logarithm and the K-exponential. We have written some proofs in Appendix A for our and reader's convenience.

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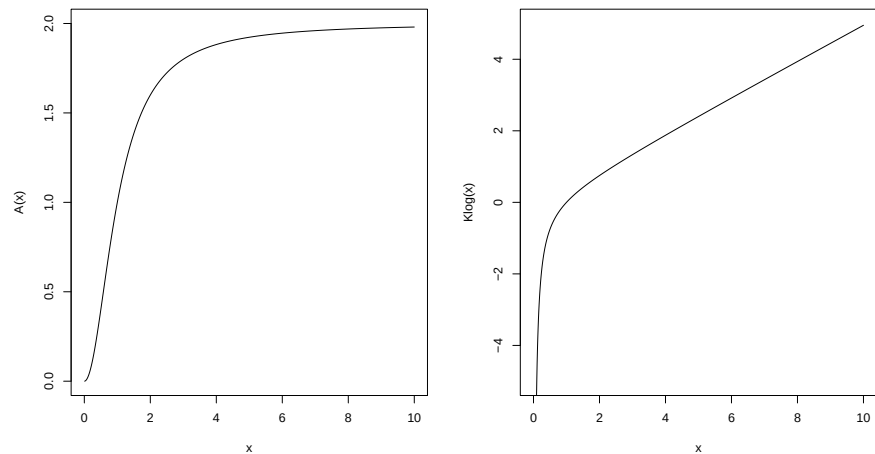


Figure 1. Naudt's function and Kaniadakis' logarithm.

$$\log_K\left(\frac{1}{x}\right) = -\log_K(x), \quad \exp_K(y) \exp_K(-y) = 1. \quad (3)$$

$$\frac{d}{dy} \exp_K(y) = A(\exp_K(y)) = \frac{1}{\sqrt{1+y^2}} \exp_K(y) \quad (4)$$

$$\exp_K(x) \exp_K(y) = \exp_K\left(x\sqrt{1+y^2} + y\sqrt{1+x^2}\right) \quad (5)$$

$$\log_K(x_2) - \log_K(x_1) < \frac{1}{A(x_1)}(x_2 - x_1), \quad x_1 \neq x_2 \quad (6)$$

$$\exp_K(y_2) - \exp_K(y_1) > A(\exp_K(y_1))(y_2 - y_1), \quad y_1 \neq y_2 \quad (7)$$

1.2. K -exponential form of a positive probability density

If Ω is a finite sample space, the probability simplex on Ω is $\mathcal{P}(\Omega)$ and the open probability simplex is $\mathcal{E}(\Omega)$.

For all $p \in \mathcal{E}(\Omega)$, the function $A \circ p$, A as in eq. (1), is strictly positive and provides a positive weight on Ω . It is proportional, but usually not equal, to a probability function,

$$\tilde{p} \propto A \circ p, \quad \text{namely,} \quad \tilde{p}(x) = \frac{A(p(x))}{\sum_{x \in \Omega} A(p(x))}.$$

We will also write $\tilde{p} = A \circ p / \overline{A \circ p}$. The mapping $\mathcal{E}(\Omega) \ni p \mapsto \tilde{p} \in \mathcal{E}(\Omega)$ is the *escort* mapping, see [3]. See [5, §3.1] for a discussion of injectivity and surjectivity. We introduce a symbol for the escort expectation. $\tilde{\mathbb{E}}_p[u] = \sum_{x \in \Omega} u(x) \tilde{p}(x)$.

For $p, q \in \mathcal{E}(\Omega)$, we define the *Kaniadakis divergence* as

$$\begin{aligned} \tilde{D}(p|q) &= \tilde{\mathbb{E}}_p[\log_K(p) - \log_K(q)] \\ &= \sum_{x \in \Omega} (\log_K(p) - \log_K(q)) A \circ p / \overline{A \circ p} \\ &= \sum_{x \in \Omega} (\log_K(p) - \log_K(q)) \tilde{p}(x) \end{aligned} \quad (8)$$

Clearly, $\tilde{D}(p|p) = 0$. If $p \neq q$, from the concavity in eq. (6),

$$\begin{aligned}
\tilde{D}(p|q) &= (\overline{A \circ p})^{-1} \sum_{x \in \Omega} A(p(x))(\log_K(p(x)) - \log_K(q(x))) \\
&> (\overline{A \circ p})^{-1} \sum_{x \in \Omega} (p(x) - q(x)) \\
&= 0.
\end{aligned}$$

Fix $p \in \mathcal{E}(\Omega)$. For all $q \in \mathcal{E}(\Omega)$, define

$$s_p(q) = (\log_K(q) - \log_K(p)) + \tilde{D}(p|q) = (\log_K(q) - \log_K(p)) - \tilde{\mathbb{E}}_p[(\log_K(q) - \log_K(p))],$$

then, for $u = s_p(q)$,

$$q = \exp_K(u - \tilde{D}(p|q) + \log_K(p)), \quad \tilde{\mathbb{E}}_p[u] = 0. \quad (9)$$

Conversely, for all $p \in \mathcal{E}(\Omega)$, if u is a random variable such that $\tilde{\mathbb{E}}_p[u] = 0$, the real function

$$\mathbb{R}_+ \ni \psi \mapsto \sum_{x \in \Omega} \exp_K(u(x) - \psi + \log_K(p(x)))$$

is continuous, goes to 0 as $\psi \rightarrow \infty$, and, for $\psi = 0$, takes a value larger than 1 because of eq. (7):

$$\begin{aligned}
\sum_{x \in \Omega} \exp_K(u(x) + \log_K(p(x))) &> \sum_{x \in \Omega} \exp_K(\log_K(p(x))) + \sum_{x \in \Omega} A(\exp_K(\log_K(p(x))))u(x) \\
&= \sum_{x \in \Omega} p(x) + \sum_{x \in \Omega} A(p(x))u(x) \\
&= 1.
\end{aligned}$$

In conclusion, there exists a function,

$$K_p: S_p = \left\{ u \in L(\Omega) \mid \tilde{\mathbb{E}}_p[u] = 0 \right\} \ni u \mapsto K_p(u) \geq 0,$$

and $K_p(u) > 0$ provided $u \neq 0$, such that

$$q = \exp_K(u - K_p(u) + \log_K(p)) \in \mathcal{E}(\Omega),$$

hence $K_p(u) = \tilde{D}(p|q)$. The mapping

$$s_p: \mathcal{E}(\Omega) \rightarrow S_p$$

is a bijection with inverse

$$e_p: S_p \ni u \mapsto \exp_K(u - K_p(u) + \log_K(p)) \in \mathcal{E}(\Omega). \quad (10)$$

1.3. Properties of the cumulant function K_p .

Let us compute the derivatives of the function K_p . We use a square bracket notation for the direction,

$$dK_p(u)[h] = \lim_{\theta \rightarrow 0} \theta^{-1} (K_p(u + \theta h) - K_p(u)) = \left. \frac{d}{d\theta} K_p(u + \theta h) \right|_{\theta=0}.$$

From eq. (10),

$$\begin{aligned}
0 &= \frac{d}{d\theta} \sum_{x \in \Omega} \exp_K(u(x) + \theta h(x) - K_p(u + \theta h) + \log_K(p(x))) \Big|_{\theta=0} \\
&= \sum_{x \in \Omega} A(\exp_K(u(x) + \theta h(x) - K_p(u + \theta h) + \log_K(p(x))))(h(x) - \frac{d}{dt} K_p(u + \theta h)) \Big|_{\theta=0} \\
&= \sum_{x \in \Omega} A(\exp_K(u(x) - K_p(u) + \log_K(p(x))))(h(x) - dK_p(u)[h]) \\
&= \sum_{x \in \Omega} A(e_p(u))(h - dK_p(u)[h]) .
\end{aligned}$$

It follows that for each $p \in \mathcal{E}(\Omega)$, and $u, h \in S_p$, it holds

$$dK_p(u)[h] = \tilde{\mathbb{E}}_q[h] , \quad \text{where } q = e_p(u) \text{ see eq. (10)} . \quad (11)$$

Remark 1. Similarly, one should compute $d^2 K_p$ and the convex conjugate of K_p . See below for the duality and see [6], [7], and M. S. notes.

1.4. Bibliographical notes

Kaniadakis logarithm and exponential were first introduced in Kaniadakis [8]; Kaniadakis [9]. The application to IG used here appeared in Pistone [10]; Montrucchio and Pistone [11]; Montrucchio and Pistone [5]. These papers discuss both the finite state space and the general state space.

2. Affine space

Remark 2. The non-parametric affine geometry of the open probability simplex is a variation of the standard case; see [7] and MS notes. The main difference is the substitution of the expectation with the escort expectation. Some details remain for the moment unsettled.

2.1. Statistical bundle

The statistical bundle is an expression of the tangent space of $\mathcal{E}(\Omega)$ as a dually flat affine statistical manifold in the sense of [12].

The statistical bundle $S\mathcal{E}(\Omega)$ and each fiber $S_q\mathcal{E}(\Omega)$ are defined by

$$S\mathcal{E}(\Omega) = \left\{ (q, v) \mid q \in \mathcal{E}(\Omega), \tilde{\mathbb{E}}_q[v] = 0 \right\} , \quad (12)$$

$$S_q\mathcal{E}(\Omega) = \left\{ v \in L(\Omega) \mid \tilde{\mathbb{E}}_q[v] = 0 \right\} , \quad q \in \mathcal{E}(\Omega) . \quad (13)$$

In our setup, each fibre is a finite-dimensional vector space and can be identified with its dual. However, it is convenient to distinguish the two statistical bundles. The previous one is called *exponential* statistical bundle, while the *mixture* statistical bundle is

$$^*S\mathcal{E}(\Omega) = \left\{ (q, v) \mid q \in \mathcal{E}(\Omega), \tilde{\mathbb{E}}_q[v] = 0 \right\} , \quad (14)$$

$$^*S_q\mathcal{E}(\Omega) = \left\{ v \in L(\Omega) \mid \tilde{\mathbb{E}}_q[v] = 0 \right\} , \quad q \in \mathcal{E}(\Omega) . \quad (15)$$

For each couple $p, q \in \mathcal{E}(\Omega)$, the mapping

$${}^e\mathbb{U}_p^q: S_p\mathcal{E}(\Omega) \ni v \mapsto v - \tilde{\mathbb{E}}_q[v] \in S_q\mathcal{E}(\Omega) \quad (16)$$

is a bijection. The ${}^e\mathbb{U}_p^p$ is the identity mapping, and

$${}^e\mathbb{U}_q^r {}^e\mathbb{U}_p^q = {}^e\mathbb{U}_p^r ,$$

The co-cycle of mappings $({}^e\mathbb{U}_p^q)_{p,q}$ is the *exponential parallel transport* of the exponential statistical bundle.

The mapping defined for all $p \in \mathcal{E}(\Omega)$, $v \in {}^*S_p \mathcal{E}(\Omega)$, and $w \in S_p \mathcal{E}(\Omega)$ by

$$g: (p, v, w) \mapsto g_p(v, w) = \langle v, w \rangle_{A \circ p} = \sum_{x \in \Omega} A(p(x)) u(x) v(x) = \overline{A \circ p} \tilde{\mathbb{E}}_p[vw]$$

provides a duality between the fibres of $S \mathcal{E}(\Omega)$ and ${}^*S \mathcal{E}(\Omega)$.

Let us compute the dual of the exponential transport. For $p, q \in \mathcal{E}(\Omega)$, $v \in {}^*S_q \mathcal{E}(\Omega)$, and $w \in S_p \mathcal{E}(\Omega)$,

$$\begin{aligned} \langle v, {}^e\mathbb{U}_p^q w \rangle_{A \circ q} &= \sum_{x \in \Omega} A \circ q v(w - \tilde{\mathbb{E}}_q[w]) \\ &= \sum_{x \in \Omega} A \circ q v w - \tilde{\mathbb{E}}_q[w] \sum A \circ q v \\ &= \sum_{x \in \Omega} A \circ q v w \\ &= \sum_{x \in \Omega} A \circ p \left(\frac{A \circ q}{A \circ p} u \right) w \\ &= \left\langle \frac{A \circ q}{A \circ p} u, w \right\rangle_{A \circ p}. \end{aligned}$$

Now, $\frac{A \circ q}{A \circ p} v \in {}^*S_p \mathcal{E}(\Omega)$, hence the dual of the exponential transport is the *mixture transport*,

$${}^*({}^e\mathbb{U}_q^p) v = {}^m\mathbb{U}_q^p v = \frac{A \circ q}{A \circ p} v. \quad (17)$$

2.2. Velocity and auto-parallel curves

The following computation is a version of the original argument about Fisher's score. Let $t \mapsto q(t) \in \mathcal{E}(\Omega)$ be a one-dimensional parametric model, a curve in geometric language. We assume it is smooth as a mapping in the vector space $L(\Omega)$. For each random variable $f \in L(\Omega)$,

$$\begin{aligned} \frac{d}{dt} \mathbb{E}_{q(t)}[f] &= \sum_{x \in \Omega} f(x) \dot{q}(x; t) \\ &= \sum_{x \in \Omega} f(x) \frac{\dot{q}(x; t)}{A(q(x; t))} A(q(x; t)) \\ &= \sum_{x \in \Omega} f(x) \frac{d}{dt} \log_K(q(x; t) A(q(x; t))) \\ &= \sum_{x \in \Omega} A \circ q(t) (f - \mathbb{E}_{\tilde{q}(t)}[f]) \frac{d}{dt} \log_K(q(t)) \\ &= \left\langle f - \mathbb{E}_{\tilde{q}(t)}[f], \frac{d}{dt} \log_K(q(t)) \right\rangle_{A \circ q(t)}. \end{aligned}$$

If we define the *velocity* of the curve to be

$$\dot{q}(t) = \frac{d}{dt} \log_K(q(t)) = \frac{\dot{q}(t)}{A(q(t))}. \quad (18)$$

we can check that $\dot{q}(t) \in S_{q(t)} \mathcal{E}(\Omega)$ and $(f - \mathbb{E}_{\tilde{q}(t)}[f]) \in {}^*S_{q(t)} \mathcal{E}(\Omega)$. And symmetrically.

The *Cramer-Rao bound* is

$$\begin{aligned}
\left(\frac{d}{dt} \mathbb{E}_{q(t)}[f] \right)^2 &= \overline{A \circ q(t)}^2 \left\langle f - \mathbb{E}_{\tilde{q}(t)}[f], \frac{d}{dt} \log_K(q(t)) \right\rangle_{\tilde{q}(t)} \\
&\leq \overline{A \circ q(t)}^2 \mathbb{E}_{\tilde{q}(t)} \left[(f - \mathbb{E}_{\tilde{q}(t)}[f])^2 \right] \mathbb{E}_{\tilde{q}(t)} \left[\left(\frac{\dot{q}(t)}{A \circ q(t)} \right)^2 \right] \\
&= \sum_{x \in \Omega} A \circ q(t) (f - \tilde{\mathbb{E}}_q[f])^2 \sum \frac{\dot{q}(t)^2}{A \circ q(t)} .
\end{aligned}$$

Remark 3. Notice that the variation computed with the escort probability function, namely $f - \tilde{\mathbb{E}}_q[f]$ appears in ?? as a gradient of the expectation $f \mapsto \mathbb{E}_q[f]$.

A curve $t \mapsto q(t)$ is auto-parallel for the mixture transport if

$$\mathfrak{m}\mathbb{U}_{q(t)}^{q(s)} \dot{q}(t) = \dot{q}(s) .$$

For $q(0) = q_0$ and $q(1) = q_1$,

$$\frac{\dot{q}(0)}{A \circ q_0} = \frac{A \circ q(t)}{A \circ q_0} \frac{\dot{q}(t)}{A \circ q(t)} = \frac{1}{A \circ q_0} \dot{q}(t) ,$$

so that

$$q(t) = q_0 + t(q_1 - q_0) .$$

The auto-parallel curves for the mixture transport are the ordinary geodesics of the simplex.

Let us compute the auto-parallel curves for the exponential transport,

$$\mathfrak{e}\mathbb{U}_{q(t)}^{q(s)} \dot{q}(t) = \dot{q}(s) .$$

For $q(0) = q_0$ and $q(1) = q_1$,

$$\dot{q}(0) = \dot{q}(t) - \tilde{\mathbb{E}}_{q_0}[\dot{q}(t)] = \frac{d}{dt} \log_K(q(t)) - \frac{1}{A \circ q_0} \sum A \circ q_0 \frac{d}{dt} \log_K(q(t)) ,$$

so that, for some function ψ

$$\log_K(q(t)) = \log_K(q_0) + t\dot{q}(0) - \psi(t) .$$

Comparing with eq. (10), we have that the auto-parallel curve for the exponential transport is

$$q(t) = \exp_K(t\dot{q}(0) - K_{q_0}(t\dot{q}(0)) + \log_K(q_0)) .$$

As seen above,

$$\dot{q}(0) = \log_K(q_1) - \log_K(q_0) - \tilde{\mathbb{E}}_{q_0}[\log_K(q_1) - \log_K(q_0)] = \log_K(q_0) + \tilde{D}(q_0|q_1) .$$

2.3. Displacement

The machinery introduced above allows for the explicit definition of the affine structure in the sense of Weyl [13].

DEFINITIONS

Mixture displacement

$$\eta_p(q) = \frac{q - p}{A \circ p} \quad (19)$$

Exponential displacement

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$$s_p(q) = (\log_K(q) - \log_K(p)) - \tilde{\mathbb{E}}_p[(\log_K(q) - \log_K(p))] \quad (20)$$

Both define affine coordinates (check it)

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DEFINITION OF an AFFINE ATLAS on the statistical bundle

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2.4. *Velocity and acceleration*

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2.5. *Gradient*

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2.6. *Bibliographical notes*

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3. Compositional data

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The Center for Risk Management at the University of Lausanne provides systemic risk assessments for European financial institutions, which we used in our empirical study using the above Kaniadakis methods. The dataset enables the determination of SRISK country-level values, a market-based systemic risk indicator first proposed in [14,15] and most recently examined in [16].

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The characteristics of SRISK are popular in the literature, and particularly SRISK is used to recognize weak institutions and countries with a systemwide impact before a crisis occurs [17] and can help forecast real sector performance [18].

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This literature has mainly focused on the absolute values of SRISK. In this work, we focus on implementing the Kaniadakis methods to see the different European countries as a part of compositional data. We developed work started in [19], where they first introduced compositional data analysis to examine the distribution of relative contributions to SRISK connected with key European nations from 2008 to 2021.

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3.1. Description of Data

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In this research, we take the sample of ten European economies (Belgium, Denmark, France, Germany, Greece, Italy, Netherlands, Spain, Switzerland, UK) with annual SRISK measurements collected at the end of December for 2008–2021. Every number is stated in billions of euros. Like most CoDa method applications, the sample does not cover Europe. Therefore, the 10 components that make up our SRISK compositions are just a portion of all those that may be used. CoDa analysis, however, is predicated on the basic notion of sub-compositional coherence, which ensures that a compositional study conducted on a subset of components is consistent with the same analysis performed on the entire composition.

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3.2. Data Analysis

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3.2.1. Kaniadakis divergence

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First, we compute the Kaniadakis divergence defined in equation (8). Each point in the figure (2) is the Kaniadakis divergence of the year in the i^{th} row concerning the j^{th} column. The Kaniadakis divergence is the difference between two probability distributions and is also defined as strictly positive in the probability simplex. The resultant distance is a statistical distance when the points are understood as probability distributions. For example, the Kaniadakis divergence between 2020 and 2021 $\bar{K}\bar{D}(2008, 2009) = 0.14$. The year 2009 deviates significantly from other years.

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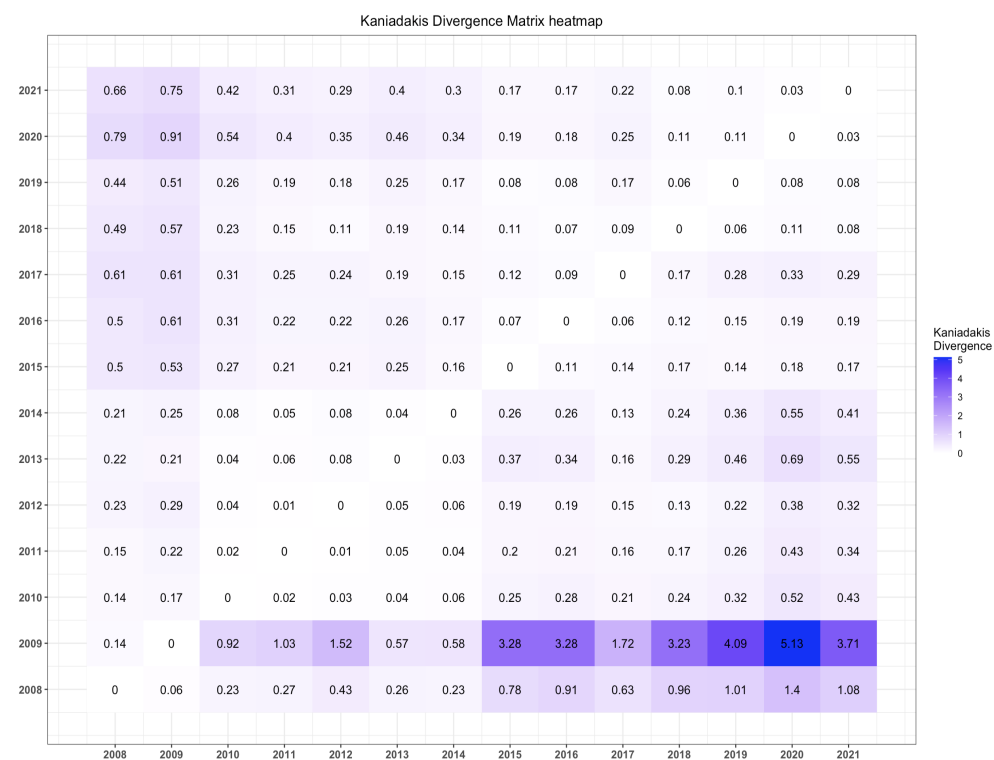


Figure 2. Kaniadakis Divergence on Compositional Data

3.2.2. Mixture Displacement

The equation (19) calculates the mixture displacement. From the figure (3), the mixture displacement for Greece and Spain is very high for all the years. The value for Spain in 2009 was less than zero—the only negative value for Spain. On the contrary, all other countries do not have too many high values.



Figure 3. Mixture Displacement on Compositional Data by taking 2008 as reference

3.2.3. Exponential Displacement

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The equation (20) computes the exponential displacement. Figure (4) is the empirical result of equation (20). Like mixture displacement, while calculating exponential displacement, we can see that Spain and Greece have higher displacement than other European countries. The value for Spain in the year 2009 is meager.

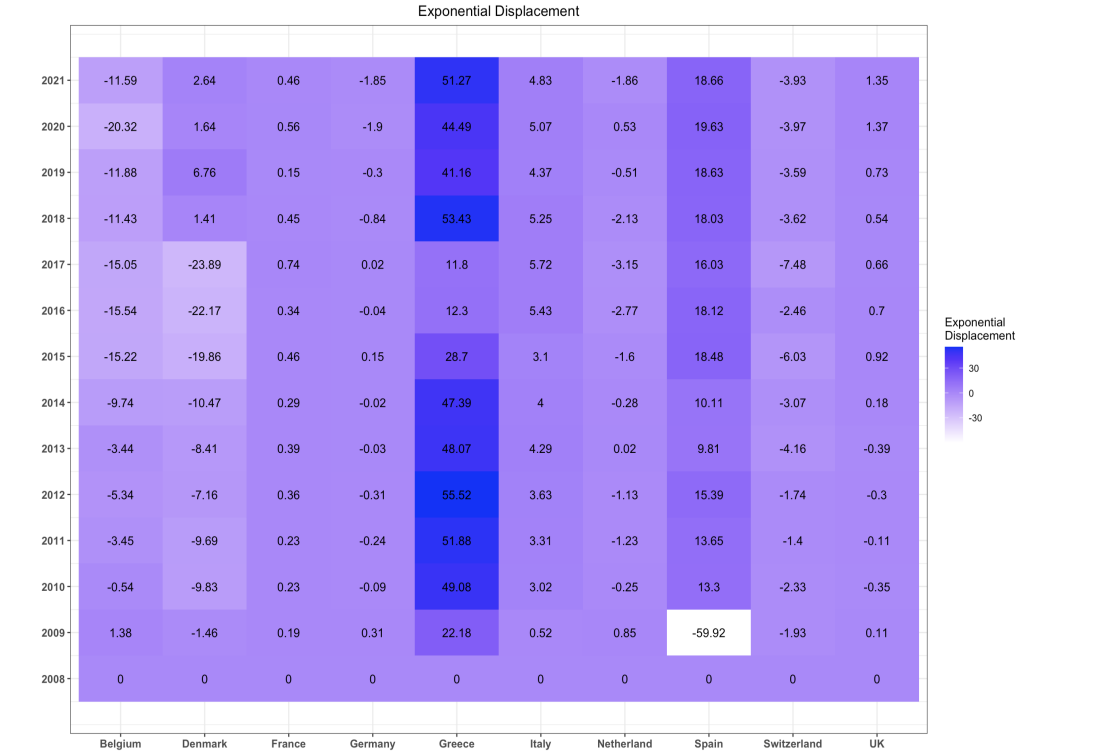


Figure 4. Exponential Displacement on Compositional Data by taking 2008 as reference

The computation of the affine displacement is a network computation; see the R suite or Python.

The computation of the exponential is a network equation.

FIND A GOOD EXAMPLE FOR ILLUSTRATION AND TEST

This appendix contains extra material, variants, and tentative redactions.

Appendix A. Some Proofs

Proof of eq. (5).

Compute $\log_K(\exp_K(x) \exp_K(y))$ using the algebraic expression of $\log_K$ and $\exp_K$.

Appendix B. Variations

Appendix B.1. K -exponential form of a positive probability density

If Ω is a finite sample space, the probability simplex on Ω is $\mathcal{P}(\Omega)$, and the open probability simplex is $\mathcal{E}(\Omega)$.

For all $p \in \mathcal{E}(\Omega)$, the function $A \circ p$, A as in eq. (1), is strictly positive and proportional to a probability function $\tilde{p}$,

$$\tilde{p} \propto A \circ p, \quad \text{namely,} \quad \tilde{p}(x) = \frac{A(p(x))}{\sum_x A(p(x))}.$$

The mapping

$$\mathcal{E}(\Omega) \ni p \mapsto \tilde{p} \in \mathcal{E}(\Omega)$$

is called the *escort* mapping, see [3]. We introduce a symbol for the escort expectation,

$$\tilde{\mathbb{E}}_p[u] = \mathbb{E}_{\tilde{p}}[u] = \frac{\sum_x u(x) A(p(x))}{\sum_x A(p(x))}.$$

For $p, q \in \mathcal{E}(\Omega)$ write q as a deformed exponential model centred at p , we define the *Kaniadakis divergence* as $\tilde{\mathbb{E}}_p[\log_K(p) - \log_K(q)] = \tilde{D}(p|q)$. Clearly, $\tilde{D}(p|p) = 0$. If $p \neq q$, from eq. (6),

$$\tilde{D}(p|q) \propto \sum_x A(p(x))(\log_K(p) - \log_K(q)) > \sum_x (p(x) - q(x)) = 0.$$

Fix $p \in \mathcal{E}(\Omega)$. For all $q \in \mathcal{E}(\Omega)$, define

$$s_p(q) = (\log_K(q) - \log_K(p)) + \tilde{D}(p|q) = (\log_K(q) - \log_K(p)) - \tilde{\mathbb{E}}_p[(\log_K(q) - \log_K(p))].$$

If $u = s_p(u)$, then

$$q = \exp_K(u - \tilde{D}(p|q) + \log_K(p)), \quad \tilde{\mathbb{E}}_p[u] = 0. \quad (\text{A1})$$

Conversely, for all $p \in \mathcal{E}(\Omega)$, if u is a random variable such that $\tilde{\mathbb{E}}_p[u] = 0$, the real function

$$\mathbb{R}_+ \ni \psi \mapsto \sum_x \exp_K(u(x) - \psi + \log_K(p(x)))$$

is continuous, goes to 0 as $\psi \rightarrow \infty$, and, for $\psi = 0$, takes a value larger than 1 because of eq. (7):

$$\begin{aligned} \sum_x \exp_K(u(x) + \log_K(p(x))) &> \\ \sum_x \exp_K(\log_K(p(x))) + \sum_x A(\exp_K(\log_K(p(x))))u(x) &= \\ \sum_x p(x) + \sum_x A(p(x))u(x) &= 1. \end{aligned}$$

In conclusion, there exist a function

$$K_p: S_p = \left\{ u \in L(\Omega) \mid \tilde{\mathbb{E}}_p[u] = 0 \right\} \ni u \mapsto K_p(u) > 0,$$

such that

$$q = \exp_K(u - K_p(u) + \log_K(p)) \in \mathcal{E}(\Omega),$$

hence $K_p(u) = \tilde{D}(p|q)$. The mapping

$$s_p: \mathcal{E}(\Omega) \rightarrow B_p$$

is a bijection with inverse

$$e_p(u) = \exp_K(u - K_p(u) + \log_K(p)) \quad (\text{A2})$$

Appendix B.2. Properties of the cumulant function K_p .

Let us compute the derivatives of the function K_p . We use the notation

$$dK_p(u)[h] = \lim_{\theta \rightarrow 0} \theta^{-1} (K_p(u + \theta h) - K_p(u)) = \left. \frac{d}{d\theta} K_p(u + \theta h) \right|_{\theta=0}.$$

From eq. (10),

$$\begin{aligned} 0 &= \left. \frac{d}{d\theta} \sum_x \exp_K(u(x) + \theta h(x) - K_p(u + \theta h) + \log_K(p(x))) \right|_{\theta=0} = \\ &= \sum_x A(\exp_K(u(x) + \theta h(x) - K_p(u + \theta h) + \log_K(p(x))))(h(x) - \frac{d}{dt} K_p(u + \theta h)) \Big|_{\theta=0} = \\ &= \sum_x A(\exp_K(u(x) - K_p(u) + \log_K(p(x))))(h(x) - dK_p(u)[h]). \end{aligned}$$

It follows that for each $p \in \mathcal{E}(\Omega)$, $u, h \in B_p$, it holds

$$dK_p(u)[h] = \tilde{\mathbb{E}}_q[h], \quad \text{where } q = e_p(u) \text{ see eq. (10)}. \quad (\text{A3})$$

Similarly, one should compute $d^2 K_p$ and the convex conjugate of K_p . See [6], [7], and M. S. notes.

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